



Task 03: Security Operations Center (SOC)- Log Analysis, Threat Intelligence, and Incident Response Workflow

Cybersecurity Internship – CyArt

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Introduction

This task focuses on advanced Security Operations Center (SOC) operations, including log analysis, threat intelligence integration, incident escalation, alert triage, evidence handling, and full incident response simulation. The purpose of this task is to develop practical skills required to detect, analyze, and respond to real-world cyber security incidents using SOC tools and methodologies.

In modern cybersecurity environments, attackers use complex techniques that cannot be detected through simple monitoring. Therefore, SOC analysts must correlate logs from multiple sources such as firewalls, endpoints, and applications to identify attack patterns and anomalies. Advanced log analysis helps in detecting suspicious activities like unauthorized logins, unusual data transfers, and abnormal system behavior.

Threat intelligence integration plays a key role in enhancing detection capabilities by providing real-time information about known malicious IPs, domains, hashes, and attacker tactics. By integrating threat feeds into SIEM tools like Wazuh or Elastic Security, security teams can automatically enrich alerts and quickly identify whether an activity is malicious or legitimate.

Incident escalation workflows ensure that security incidents are properly handled based on severity and complexity. Low-level alerts are handled by Tier 1 analysts, while complex and critical incidents are escalated to Tier 2 or Tier 3 teams for deeper investigation. Proper communication and structured reporting such as SITREPs help maintain clarity during incident handling.



Alert triage with threat intelligence helps SOC analysts validate security alerts using external tools like VirusTotal and AlienVault OTX. This process reduces false positives and ensures that only genuine threats are escalated for response. Finally, a full SOC workflow simulation demonstrates the complete lifecycle of a cyber incident—from attack simulation using tools like Metasploit, to detection in Wazuh, response and containment using tools like CrowdSec, escalation via TheHive, and final reporting using structured documentation. This ensures a practical understanding of real-world SOC operations.

1. Advanced Log Analysis

Introduction

Advanced Log Analysis is the process of examining and correlating logs from multiple sources such as firewalls, endpoints, and applications to detect complex cyber threats. In a SOC environment, it helps analysts identify attack patterns, detect anomalies, and reduce false positives.

It is important for:

- detecting advanced attacks
- identifying attack patterns
- correlating multiple log sources
- improving incident detection
- reducing false alerts

The main goal of advanced log analysis is to understand the relationship between different log events and identify suspicious behavior.

Core Concepts

1. Log Correlation

Log correlation means combining logs from different sources to identify related events.

Example:

- Failed login attempts (Event ID 4625)
- Followed by unusual outbound traffic



This may indicate a brute-force attack or compromised account.

2. Anomaly Detection

Anomaly detection identifies unusual behavior in system activity.

Examples:

- logins at unusual times
- high data transfer volume
- multiple failed login attempts

3. Log Enrichment

Log enrichment adds extra context to logs to improve analysis.

Examples:

- IP geolocation
- user role information
- device details

Practical Knowledge

Practical 1: Log Correlation

Analyze login and network logs together:

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Failed login attempt
2025-08-18 12:01:00	4625	192.168.1.100	8.8.4.4	Suspicious DNS request

Observation:

Repeated failed logins followed by external DNS requests may indicate malicious activity.



Practical 2: Anomaly Detection Rule

Create detection rule:

Condition:

bytes_out > 1MB in 1 minute

Test Case:

Simulate large file transfer and verify alert generation.

Practical 3: Log Enrichment

Use GeoIP to enrich IP address:

Example:

IP: 8.8.8.8 → Location: United States

Findings:

- Identify unusual foreign IP logins
- Detect suspicious access locations

Tools Used

- Elastic Security
- Security Onion
- Wazuh
- Google Sheets
- GeoIP tools

Skills Learned

- log correlation
- anomaly detection



- log enrichment
- threat identification
- SOC log analysis workflow

2. Threat Intelligence Integration

Introduction

Threat Intelligence Integration is the process of using external and internal threat data to improve detection, analysis, and response in a SOC environment. It helps security analysts identify known malicious indicators and understand attacker behavior.

It is important for:

- identifying known malicious IPs, domains, and hashes
- improving alert accuracy
- supporting proactive threat hunting
- enhancing incident response
- reducing false positives

The main goal is to integrate threat intelligence into security tools like SIEM to automatically enrich alerts and improve decision-making.

Core Concepts

1. Threat Intelligence Types

- IOCs (Indicators of Compromise): malicious IPs, domains, hashes
- TTPs (Tactics, Techniques, Procedures): attacker behavior patterns
- Threat Feeds: external data sources like AlienVault OTX

2. STIX/TAXII Standards

- STIX: format for sharing threat data
- TAXII: protocol for exchanging threat intelligence



3. Threat Intelligence in SOC

Threat intelligence is used to:

- match alerts with known threats
- enrich logs with context
- automate detection in SIEM

Example:

Suspicious IP detected → matched with known C2 server in threat feed

Practical Knowledge

Practical 1: Threat Feed Import

Import AlienVault OTX feed into Wazuh and test IOC matching.

Test Case:

- Mock IP: 192.168.1.100
- Check if IP exists in threat database

Observation:

If matched → system marks IP as malicious

Practical 2: Alert Enrichment

Enrich Wazuh alert using threat intelligence:

Alert ID	IP	Reputation	Notes
-----	-----	-----	-----
003	192.168.1.100	Malicious (OTX)	Linked to C2 server

Observation:

Alerts become more meaningful with threat context.

Practical 3: Threat Hunting



Search for suspicious user activity:

Query Example:

user.name != "system"

Goal:

Identify possible misuse of valid accounts (MITRE T1078)

Observation:

Detect unauthorized or unusual user activity

Tools Used

- Wazuh
- AlienVault OTX
- MITRE ATT&CK
- STIX/TAXII feeds
- TheHive

Skills Learned

- threat intelligence integration
 - IOC analysis
 - alert enrichment
 - proactive threat hunting
 - MITRE ATT&CK mapping
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3. Incident Escalation Practice

Introduction



Incident Escalation Practice is the process of transferring security incidents to higher SOC tiers based on severity, complexity, and impact. It ensures that critical incidents are handled by skilled analysts quickly and efficiently.

It is important for:

- handling high-severity security incidents
- ensuring proper incident ownership
- improving response time
- structured communication in SOC teams
- efficient workflow management

The main goal is to escalate incidents properly so that they are resolved by the right team at the right time.

Core Concepts

1. SOC Tier Structure

- Tier 1: Initial triage and alert monitoring
- Tier 2: Detailed investigation and analysis
- Tier 3: Advanced threat analysis and response

2. Escalation Criteria

Incidents are escalated based on:

- severity level (High/Critical)
- unknown or complex attack patterns
- confirmed security breaches
- repeated suspicious activity

3. Communication in Escalation

- SITREP (Situation Report): incident summary report
 - escalation notes: technical details for higher tiers
 - stakeholder updates: non-technical summaries
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Practical Knowledge

Practical 1: Escalation Simulation

Create a case in TheHive for a High-priority alert.

Scenario:

Unauthorized access detected on server

Escalation Summary (100 words):

A suspicious login attempt was detected on Server-Y from IP 192.168.1.200. Multiple failed authentication attempts followed by a successful login indicate possible credential compromise. The incident has been classified as High severity due to potential unauthorized access to sensitive data. Initial analysis shows MITRE technique T1078 (Valid Accounts). The affected system has been isolated and logs are being collected for further investigation. The case has been escalated to Tier 2 for deeper analysis and containment actions.

Practical 2: SITREP Draft

Title: Unauthorized Access on Server-Y

Summary:

Incident detected at 2025-08-18 13:00 from IP 192.168.1.200. Unauthorized access attempts observed.

Actions Taken:

- Server isolated from network
- Logs collected for investigation
- Incident escalated to Tier 2

Observation:

Provides clear incident overview for stakeholders.

Practical 3: Workflow Automation Concept

Create a simple SOAR workflow:



Trigger:

High severity alert detected

Action:

Automatically assign incident to Tier 2 analyst

Observation:

Improves response time and reduces manual workload

Tools Used

- TheHive
- Splunk SOAR (Phantom)
- Wazuh
- Google Docs
- SOC ticketing systems

Skills Learned

- incident escalation process
 - SOC tier structure
 - SITREP writing
 - incident communication
 - workflow automation
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4. Alert Triage with Threat Intelligence

Introduction

Alert Triage with Threat Intelligence is the process of analyzing security alerts and validating them using external threat intelligence sources. It helps SOC analysts determine whether an alert is malicious, false positive, or requires escalation.



It is important for:

- validating security alerts
- reducing false positives
- prioritizing real threats
- improving incident response accuracy
- connecting alerts with known attacker activity

The main goal is to quickly assess alerts and decide the correct response using threat intelligence data.

Core Concepts

1. Alert Triage

Alert triage means classifying alerts based on severity and relevance.

Categories:

- High: confirmed malicious activity
- Medium: suspicious but not confirmed
- Low: benign or false positive

2. IOC Validation

Indicators of Compromise (IOCs) include:

- IP addresses
- file hashes
- domains
- URLs

These are verified using tools like VirusTotal and AlienVault OTX.

3. Threat Intelligence Usage

Threat intelligence helps:

- identify known malicious sources
- enrich alert context
- support decision making in SOC



Example:

Suspicious PowerShell execution → checked against threat feeds → confirmed malware behavior

Practical Knowledge

Practical 1: Alert Triage Simulation

Analyze a mock Wazuh alert:

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

Observation:

PowerShell execution is commonly used in malicious attacks and requires investigation.

Practical 2: IOC Validation

Check IP or hash using VirusTotal and OTX:

Example:

- IP: 192.168.1.101
- Result: Flagged as suspicious/malicious

Finding:

The IP is associated with known attack patterns and requires escalation.

Practical 3: Triage Decision

Decision Flow:

- If IOC is malicious → escalate to Tier 2
- If suspicious → monitor and investigate
- If clean → close alert



Observation:

Proper triage reduces unnecessary workload and improves SOC efficiency

Tools Used

- Wazuh
- VirusTotal
- AlienVault OTX
- MITRE ATT&CK
- SOC dashboards

Skills Learned

- alert triage process
 - IOC validation
 - threat intelligence usage
 - decision-making in SOC
 - incident prioritization
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5. Evidence Preservation and Analysis

Introduction

Evidence Preservation and Analysis is the process of collecting, storing, and analyzing digital evidence in a secure manner without modifying the original data. In a SOC and digital forensics environment, it ensures that evidence remains valid for investigation and legal purposes.

It is important for:

- maintaining integrity of evidence
 - supporting incident investigation
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- forensic analysis of cyber incidents
- legal and compliance requirements
- identifying attacker actions

The main goal is to preserve digital evidence in its original form and analyze it without tampering.

Core Concepts

1. Volatile Data

Volatile data is temporary data stored in RAM that disappears after shutdown.

Examples:

- running processes
- active network connections
- logged-in users

2. Evidence Integrity

Ensuring evidence is not modified during collection using:

- hashing (SHA256)
- secure storage
- controlled access

3. Chain of Custody

Chain of Custody ensures proper tracking of evidence.

It includes:

- who collected evidence
- when it was collected
- where it was stored
- who accessed it

Practical Knowledge



Practical 1: Volatile Data Collection

Command:

```
SELECT * FROM netstat
```

Observation:

- shows active network connections
- helps identify suspicious communication

Practical 2: Memory Acquisition

Command:

```
SELECT * FROM Artifact.Windows.Memory.Acquisition
```

Observation:

- captures RAM snapshot
- used for malware and process analysis

Practical 3: File Integrity Check

Generate hash:

```
sha256sum suspiciousfile.exe
```

Observation:

- ensures file has not been modified
- used for verifying authenticity of evidence

Practical 4: Evidence Documentation

Item	Description	Collected By	Date	Hash Value
-----	-----	-----	-----	-----
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	<SHA256>



Observation:

Proper documentation ensures evidence traceability.

Tools Used

- Velociraptor
- FTK Imager
- SHA256 tools
- Windows/Linux forensic utilities
- SOC forensic platforms

Skills Learned

- evidence collection
 - memory analysis
 - file integrity verification
 - chain of custody management
 - forensic documentation
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6. Capstone Project: Full SOC Workflow Simulation

Introduction

The Capstone Project is a complete SOC workflow simulation that combines all cybersecurity operations including attack simulation, detection, analysis, response, escalation, and reporting. It represents a real-world Security Operations Center (SOC) incident handling process.

It is important for:

- understanding end-to-end SOC workflow
 - simulating real cyberattacks
 - improving incident response skills

 - practicing detection and containment
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- learning reporting and escalation processes

The main goal is to simulate a real attack scenario and handle it like a professional SOC analyst.

Core Concepts

1. SOC Workflow Stages

- Attack Simulation
- Detection and Monitoring
- Triage and Analysis
- Response and Containment
- Escalation
- Reporting

2. MITRE ATT&CK Mapping

Attacks are mapped to techniques such as:

- T1210: Exploitation of Remote Services
- T1078: Valid Accounts

3. Incident Response Lifecycle

Based on NIST framework:

- Preparation
- Detection
- Containment
- Eradication
- Recovery
- Reporting

Practical Knowledge

Practical 1: Attack Simulation



Tool: Metasploit

Exploit:

use exploit/multi/samba/usermap_script

Target:

Metasploitable2 VM

Observation:

- successful exploitation indicates vulnerability in SMB service

Practical 2: Detection and Triage

Tool: Wazuh

Detected Alert:

Timestamp	Source IP	Alert Description	MITRE Technique
-----	-----	-----	-----
2025-08-18 14:00:00	192.168.1.101	Samba exploit	T1210

Observation:

Attack detected through SIEM logs and mapped to MITRE ATT&CK.

Practical 3: Response and Containment

Actions:

- isolate affected VM
- block attacker IP using CrowdSec
- verify blocking using ping test

Observation:

Containment prevents further attacker activity.

Practical 4: Escalation



Tool: TheHive

Case Summary (100 words):

A Samba service exploitation attempt was detected on Metasploitable2 VM from IP 192.168.1.101. The attack successfully exploited a known vulnerability (T1210). Immediate containment actions were taken by isolating the affected system and blocking the attacker IP using CrowdSec. Logs were collected for further investigation. Due to the severity of the incident, the case has been escalated to Tier 2 for deeper forensic analysis and impact assessment. Continuous monitoring has been enabled to detect any further malicious activity.

Practical 5: Reporting

Report (200 words summary):

A security incident was detected involving exploitation of Samba service on a vulnerable system. The attack originated from IP 192.168.1.101 and was identified through Wazuh SIEM alerts. The attacker exploited a remote service vulnerability mapped to MITRE ATT&CK technique T1210. Immediate response actions included isolating the affected machine, blocking the attacker IP using CrowdSec, and collecting forensic logs for analysis. The incident was escalated to Tier 2 analysts for deeper investigation. No lateral movement was observed after containment.

Recommendation: Patch vulnerable services, disable unnecessary SMB services, and implement continuous monitoring using SIEM tools.

Practical 6: Manager Briefing

A Samba service attack was detected on a vulnerable system. The attacker attempted unauthorized access using a known exploit. The system was immediately isolated, and the threat was contained successfully. No data loss was observed. The issue has been escalated for further investigation, and preventive security measures are being recommended.



Tools Used

- Metasploit
- Wazuh
- CrowdSec
- TheHive
- Google Docs
- Kali Linux
- Metasploitable2

Skills Learned

- full SOC workflow execution
 - attack simulation
 - incident detection and response
 - SIEM analysis
 - escalation and reporting
 - forensic investigation
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